

ADITYA KADAM

AI Engineer — LLM Applications, Agentic Systems & Production RAG

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SUMMARY

AI Engineer who ships complete production systems — model integration, multi-agent orchestration, retrieval-augmented generation, computer vision, and the backend and frontend infrastructure that ties them together. At VSky Solutions, independently designed and shipped a production RAG assistant end-to-end (retrieval, reranking, grounded generation, audit logging, and an embeddable JS widget), architected a multi-agent ERP automation pipeline spanning Outlook, document AI, and ERP write-back, and led the re-architecture of an enterprise SAP BusinessObjects metadata explorer. Comfortable choosing the right tool for a problem: an LLM agent where judgment is needed, a deterministic engine where reproducibility matters, a trained CV model where the input is visual.

TECHNICAL SKILLS

- AI / Agentic Systems:** LangGraph, LangChain, Agno, Google ADK, Gemini API, Groq, LiteLLM, prompt engineering, tool/function calling, multi-agent orchestration, structured-output validation
- RAG & Retrieval:** pgvector, ChromaDB, Qdrant, Pinecone, BM25 & hybrid retrieval, embeddings, reranking, grounded generation, citations, token-usage tracking
- Computer Vision:** YOLO, OpenCV, CNNs, VGG-16, Scikit-learn
- Backend & Data:** FastAPI, Flask, PostgreSQL, SQLAlchemy, Alembic, Pydantic v2, REST APIs, SSE streaming, Docker
- Frontend:** Vue 3, Quasar, React, TypeScript
- Practices & Tools:** Git, Pytest, n8n workflow automation, Office.js, CI/CD, Agile collaboration

PROFESSIONAL EXPERIENCE

- AI Engineer Intern — VSky Solutions** Aug 2025 – Present
- Independently designed and shipped a production **SOP RAG Assistant** end-to-end (FastAPI, Gemini, PostgreSQL/pgvector): intent classification, metadata-based document selection, vector retrieval with LLM reranking, and grounded answer generation with full retrieval-audit logging and per-call token-usage tracking.
 - Built and shipped the assistant's embeddable JavaScript widget (Shadow-DOM isolation, React/Vue/vanilla integration) that any team mounts with a single script tag, backed by an esbuild bundling pipeline with a non-interactive, CI-ready build path.
 - Architected an end-to-end ERP transaction-automation pipeline (Outlook → AI → ERP) on a multi-agent backend (LangChain, LangGraph, Agno), with a custom document-extraction pipeline for PDFs/emails and FastAPI/PostgreSQL services for agent memory and state.
 - Led a full backend and frontend re-architecture of **FalconBOExplorer**, an enterprise tool for navigating SAP BusinessObjects universe metadata: a layered FastAPI service and a 33-file modular Vue frontend, migrating the app from a JSON-file prototype to a PostgreSQL-backed, lazy-loading production tool.
 - Architected the multi-agent backend for **HR Lens** (resume-checker, email-classifier, and autofill agents) and replaced an inconsistent LLM-based candidate scorer with a deterministic, JSON-configured engine across ~14 role profiles, plus an n8n-driven three-layer skill-matching pipeline — cutting recruiter screening time from ~20 minutes to under 45 seconds per candidate.

PROJECTS

- Self-Verifying RAG Platform — Python, LangGraph, React, PostgreSQL/pgvector, Redis**
- Designed and built a production-grade RAG architecture as an explicit, fully-logged LangGraph pipeline — not an autonomous agent — with a six-outcome intention router, two-layer semantic caching, and two-way answer verification (cross-document conflict detection plus a confidence-triggered check that the answer matches its own source).
 - Engineered reliability and cost controls around it: a circuit breaker on the LLM API with in-process failover state so a cache outage can't also disable the safety fallback, a golden-dataset evaluation suite (Ragas, DeepEval) scoring retrieval and faithfulness, and per-driver cost levers across caching, token ceilings, and rate limiting.
- Job Prep Engine — Qwen2.5-3B (LoRA) + Ollama, FastAPI (SSE), React + Vite**
- Fine-tuned a Qwen2.5-3B model with a LoRA adapter and served it via Ollama behind a FastAPI SSE streaming backend, with a React + Vite frontend routing across six task types by intent; resolved CPU model-loading hangs and response-truncation issues.
- Nexus — Ollama (Qwen2.5/Llama-3), FastAPI, Kuzu, PostgreSQL, Vue 3/Quasar**
- Built a persistent multi-agent system simulating a software engineering organization, including a calibration engine that fuses verbalized and behavioral confidence scores per agent per task type, on a local Ollama + Kuzu knowledge-graph stack; authored a five-phase build plan and technical specification.
- NailCareAI — Flask, VGG-16, OpenCV**
- Built a nail-disease screening app classifying 17 conditions on a VGG-16 backbone, with a multi-architecture training pipeline, class-weighted loss, and per-run experiment tracking; reached ~88.6% top-1 accuracy.

EDUCATION

- B.Tech in Computer Science & Engineering — D.Y. Patil Agriculture and Technical University, Talsande** 2022–2026
- CGPA: 9.5 / 10.0